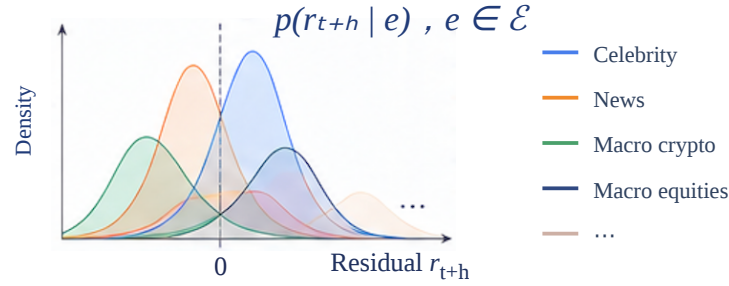


1

Event-conditioned forecast bias

Different events induce structured, multi-step forecast residual patterns.

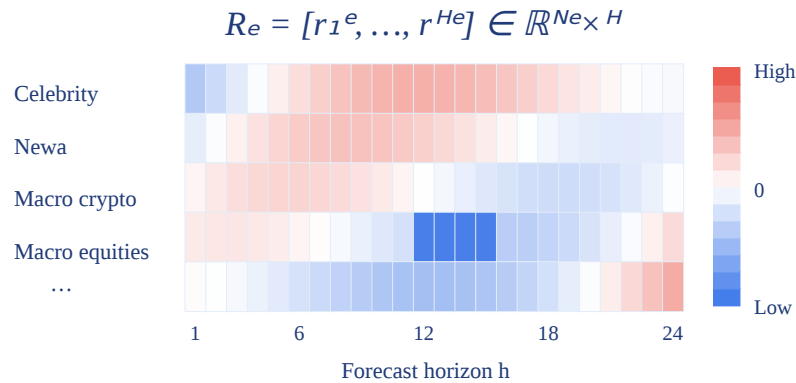
1.1 Event heterogeneity and residual distribution shift



$$r_{t+h} = y_{t+h} - \hat{y}_{t+h}, \quad p(r_{t+h} | e_i) \neq p(r_{t+h} | e_j), \quad \forall i \neq j.$$

Events induce distinct, non-stationary and structured residual distributions.

1.2 Structured multi-step residual patterns



$$r_h^e = (1/N_e) \sum_i r_{i,h}, R_e \text{ reveals event-specific temporal structures.}$$

Historical event-conditioned residuals exhibit consistent and reusable multi-step patterns.

2

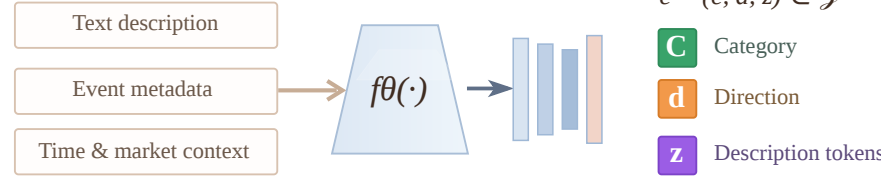
Semantic residual memory

Retrieve, transform, and calibrate historical event-conditioned residuals.

2.1 Event semantic encoding

Event e : "Trump announces new crypto reserve ..."

$$e = (c, d, z) \in \mathcal{E}$$



2.2 Event-conditioned residual memory

$$\mathcal{M} = \{(e_i, R_i)\}$$

Residual trajectories $R_i = [r_{t,1}, \dots, r_{t,H}]$

BTC

ETH

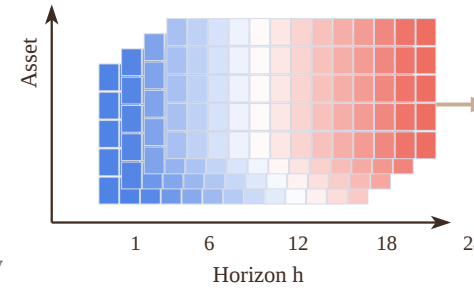
SOL

DOGE

SPX

...

Multi-asset historical memory



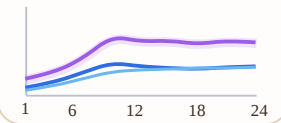
Retrieve $S(e)$

- Similarity
- Recency
- Cross-asset
- Volatility-aware

2.3 Transfer and calibration

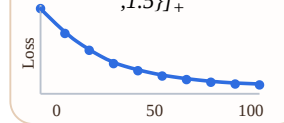
Volatility-aware transfer

$$\alpha_t(e) = D_t \cdot r^*(e)$$



Delayed-feedback calibration

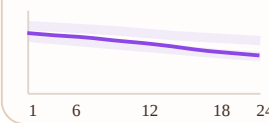
$$g_t = (n_t / (n_t + 10)) [\min\{g_{1,t}, g_{2,t}, 1.5\}]_+$$



Epochs

Calibrated correction

$$\hat{u}_t^s = \hat{u}_t^b + g_t \alpha_t$$

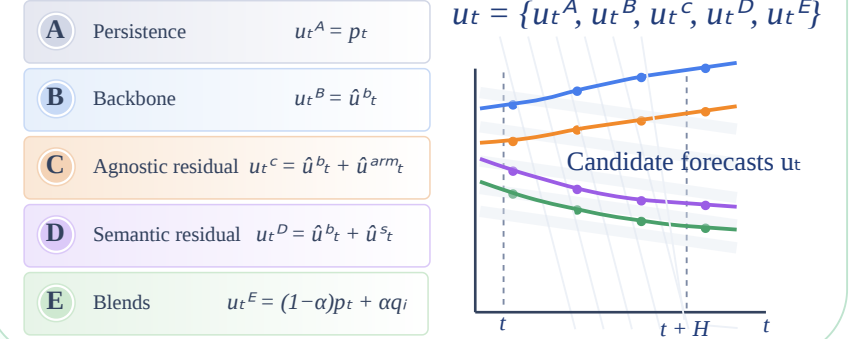


3

Reliable deployment

Route among multiple candidate forecasts with a validation-frozen reliability router.

3.1 Multiple candidate prediction paths

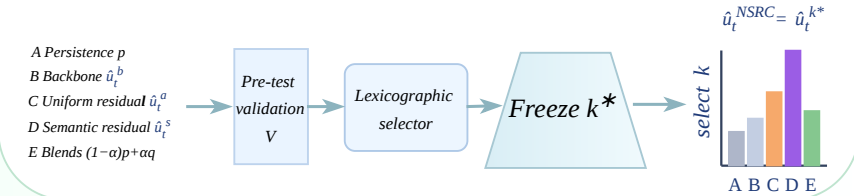


3.2 Validation-frozen reliability router

Choose the validation-best candidate path before testing; freeze the selected route throughout the test interval.

$$k^* = \text{lexargmink}(MAE_{V,k}, MSE_{V,k}, k)$$

MAE first · MSE tie-break · candidate-order tie-break



3.3 Final forecast (illustrative)

$$\hat{y}_{t+h} = \hat{y}_{t+h}^b + \hat{u}_{t+h}^{s*}$$

